

Lab 8 - 2D Arrays and Matrices (C Programs)

Q1: Classroom Marks System

```
#include <stdio.h>

int main()
{
    int marks[4][3] = {
        {80,75,90},
        {60,70,85},
        {88,92,79},
        {55,65,70}
    };

    int i,j,total;

    for(i=0;i<4;i++)
    {
        total = 0;
        for(j=0;j<3;j++)
        {
            total = total + marks[i][j];
        }
        printf("Total of student %d = %d\n",i+1,total);
    }

    for(j=0;j<3;j++)
    {
        total = 0;
        for(i=0;i<4;i++)
        {
            total = total + marks[i][j];
        }
        printf("Average of subject %d = %.2f\n",j+1,total/4.0);
    }

    return 0;
}
```

Q2: Cinema Seat Booking

```
#include <stdio.h>

int main()
{
    int seats[5][6];
    int i,j;
    int available = 0;
```

```

int maxRow = 0;
int maxBooked = 0;
int booked;

for(i=0;i<5;i++)
{
    booked = 0;
    for(j=0;j<6;j++)
    {
        scanf("%d",&seats[i][j]);

        if(seats[i][j] == 0)
            available++;
        else
            booked++;
    }

    if(booked > maxBooked)
    {
        maxBooked = booked;
        maxRow = i;
    }
}

printf("Available seats = %d\n",available);
printf("Row with max bookings = %d\n",maxRow+1);

return 0;
}

```

Q3: Temperature Monitoring

```
#include <stdio.h>
```

```

int main()
{
    float temp[7][3];
    int i,j;
    float max,sum;

    for(i=0;i<7;i++)
    {
        for(j=0;j<3;j++)
        {
            scanf("%f",&temp[i][j]);
        }
    }

    max = temp[0][0];

    for(i=0;i<7;i++)

```

```

{
    sum = 0;

    for(j=0;j<3;j++)
    {
        if(temp[i][j] > max)
            max = temp[i][j];

        sum = sum + temp[i][j];
    }

    printf("Average of day %d = %.2f\n",i+1,sum/3);
}

printf("Highest temp in week = %.2f\n",max);

return 0;
}

```

Q4: 3x3 Matrix Analysis

```
#include <stdio.h>
```

```

int main()
{
    int a[3][3],t[3][3];
    int i,j,det;

    for(i=0;i<3;i++)
        for(j=0;j<3;j++)
            scanf("%d",&a[i][j]);

    for(i=0;i<3;i++)
        for(j=0;j<3;j++)
            t[j][i] = a[i][j];

    det = a[0][0]*(a[1][1]*a[2][2]-a[1][2]*a[2][1])
        - a[0][1]*(a[1][0]*a[2][2]-a[1][2]*a[2][0])
        + a[0][2]*(a[1][0]*a[2][1]-a[1][1]*a[2][0]);

    printf("Determinant = %d\n",det);

    printf("Transpose:\n");
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
            printf("%d ",t[i][j]);
        printf("\n");
    }

    return 0;
}

```

```
}
```

Q5: Matrix Type Check

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[5][5];
```

```
    int r,c,i,j;
```

```
    int zero = 1;
```

```
    int identity = 1;
```

```
    scanf("%d %d",&r,&c);
```

```
    for(i=0;i<r;i++)
```

```
        for(j=0;j<c;j++)
```

```
            scanf("%d",&a[i][j]);
```

```
    for(i=0;i<r;i++)
```

```
    {
```

```
        for(j=0;j<c;j++)
```

```
        {
```

```
            if(a[i][j] != 0)
```

```
                zero = 0;
```

```
            if(i==j)
```

```
            {
```

```
                if(a[i][j] != 1)
```

```
                    identity = 0;
```

```
            }
```

```
            else
```

```
            {
```

```
                if(a[i][j] != 0)
```

```
                    identity = 0;
```

```
            }
```

```
        }
```

```
    }
```

```
    if(r==c)
```

```
        printf("Square Matrix\n");
```

```
    else
```

```
        printf("Rectangular Matrix\n");
```

```
    if(zero==1)
```

```
        printf("Zero Matrix\n");
```

```
    if(identity==1)
```

```
        printf("Identity Matrix\n");
```

```
    return 0;
```

```
}
```

Q6: Matrix Multiplication

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[3][3],b[3][3],res[3][3];
```

```
    int r1,c1,r2,c2;
```

```
    int i,j,k;
```

```
    scanf("%d %d",&r1,&c1);
```

```
    scanf("%d %d",&r2,&c2);
```

```
    if(c1 != r2)
```

```
    {
```

```
        printf("Multiplication not possible");
```

```
        return 0;
```

```
    }
```

```
    for(i=0;i<r1;i++)
```

```
        for(j=0;j<c1;j++)
```

```
            scanf("%d",&a[i][j]);
```

```
    for(i=0;i<r2;i++)
```

```
        for(j=0;j<c2;j++)
```

```
            scanf("%d",&b[i][j]);
```

```
    for(i=0;i<r1;i++)
```

```
    {
```

```
        for(j=0;j<c2;j++)
```

```
        {
```

```
            res[i][j] = 0;
```

```
            for(k=0;k<c1;k++)
```

```
                res[i][j] = res[i][j] + a[i][k]*b[k][j];
```

```
        }
```

```
    }
```

```
    printf("Result Matrix:\n");
```

```
    for(i=0;i<r1;i++)
```

```
    {
```

```
        for(j=0;j<c2;j++)
```

```
            printf("%d ",res[i][j]);
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

